XIV. Asymptotic Theory (GSE 518)

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1. Motivation:

$(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} f(x|\theta)$, use $\hat{\theta}$ to estimate θ_0.

1) Many times it is hard to find unbiased estimators and hence UMVUE.

- Ex: to estimate the standard deviation:

$$E(s^2) = \sigma_0^2 \quad \text{does not imply} \quad E(s) = \sigma_0$$

or say

$$E(\sqrt{s^2}) = E(s) \neq \sigma_0$$

- If we only focus on unbiased estimators, we may rule out other good estimators with slight bias but better variance.

- So we need another criteria to evaluate estimators.

Idea: Analyze the behavior of $\hat{\theta}$ as $n \to \infty$. Does $\hat{\theta}$ converge to θ_0 as $n \to \infty$?

2) $\hat{\theta}$ is a random variable $\Rightarrow$ we want to know the distribution of $\hat{\theta}$ to

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a) $E(\hat{\theta})$ and $\text{Var}(\hat{\theta})$

b) For hypothesis testing and confidence interval.

There are 3 ways to learn about Distribution of $\hat{\theta}$:

1) Derive the exact distribution of $\hat{\theta}$. (quite hard)

2) Approximate the exact distribution of $\hat{\theta}$ using simulation techniques, such as *bootstrap* (in Fall).

3) Approx. the distribution of $\hat{\theta}$ as $n \to \infty$ using *asymptotic theory*.

2. Asymptotic Theory: Introduction

(See lecture for details)

3. Convergence Concepts

3.1 Definition of Convergence in Probability

A sequence of random variables $\{X_n\}_{n=1}^{\infty}$ converges in probability to a constant c if for every $\epsilon > 0$:

$$\lim_{n\to\infty} P\left(|X_n - c| > \epsilon\right) = 0 \quad \text{or} \quad \lim_{n\to\infty} P\left(|X_n - c| < \epsilon\right) = 1$$

The probability as $n \to \infty$ that X_n is arbitrarily close to c is 1.

Notes:

- Most used concept in econometrics and is also called "weak convergence in probability".
- We write $X_n \xrightarrow{P} c$.

3.2 Definition of Convergence in Mean Square

A sequence of random variables $\{X_n\}_{n=1}^{\infty}$ converges to a constant c in mean square or quadratic mean if:

$$\lim_{n\to\infty} E\left(X_n - c\right)^2 = 0$$

We write $X_n \xrightarrow{q} c$ or $X_n \xrightarrow{m} c$.

Notes:

- Can write this as:

$$\lim_{n\to\infty} E\left[(X_n - c)^2\right] = \lim_{n\to\infty} \text{Var}(X_n) + \lim_{n\to\infty} Bias^2(X_n)$$

  where $\text{Bias}(X_n) = E(X_n) - c$.

- $\lim_{n\to\infty} \text{Var}(X_n) = 0$ and $\lim_{n\to\infty} \text{Bias}(X_n) = 0 \Rightarrow \lim_{n\to\infty} E\left[(X_n - c)^2\right] = 0 \quad \text{or} \quad X_n \xrightarrow{q} c$.

It can be shown that:

- $X_n \xrightarrow{q} c \implies X_n \xrightarrow{P} c$.
- But **NOT** $X_n \xrightarrow{P} c \not\Rightarrow X_n \xrightarrow{q} c$

3.3. Definition of consistent sequence (and estimator): A sequence of estimators $\{\hat{\theta}(x)\}_{n=1}^{\infty}$ is a consistent sequence of estimators of θ_0 if $\forall \epsilon > 0$:

$$\lim_{n\to\infty} P(|\hat{\theta} - \theta_0| > \epsilon) = 0 \quad \text{or} \quad \lim_{n\to\infty} P(|\hat{\theta} - \theta_0| < \epsilon) = 1.$$

i.e. $\hat{\theta} \xrightarrow{P} \theta_0$. We say $\hat{\theta}$ is a "consistent estimator of θ_0."

Theorem: If $\hat{\theta}$ is a sequence of estimators s.t.:

i) $\lim_{n\to\infty} \text{Bias}(\hat{\theta}) = 0$ i.e., $\lim_{n\to\infty} E(\hat{\theta}) = \theta_0$

ii) $\lim_{n\to\infty} \text{Var}(\hat{\theta}) = 0$

i.e. $\hat{\theta} \xrightarrow{q} \theta_0 \Rightarrow \hat{\theta} \xrightarrow{P} \theta_0$. $\hat{\theta}$, so is consistent for θ_0.

Note: Converse not true.

$$\hat{\theta} \xrightarrow{P} \theta_0 \not\Rightarrow \hat{\theta} \xrightarrow{q} \theta_0.$$

Example: Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu_0, 1)$ and consider $\overline{X}$ as an estimator of μ_0. Is $\overline{X}$ consistent?

Recall: $\overline{X} \sim N(\mu_0, \frac{1}{n})$.

Steps:

1. Get $E(\hat{\theta})$ and $\text{Bias}(\hat{\theta})$.
2. Get $\text{Var}(\hat{\theta})$.
3. Take $\lim_{n\to\infty} \text{Bias}(\hat{\theta})$ and $\text{Var}(\hat{\theta})$.

1)

$$E(\overline{X}) = E(X) = \mu_0 \Rightarrow \text{Bias}(\overline{X}) = 0.$$

2)

$$\text{Var}(\overline{X}) = \frac{\text{Var}(X)}{n} = \frac{1}{n}.$$

3)

$$\lim_{n\to\infty} \text{Bias}(\overline{X}) = 0$$

$$\lim_{n\to\infty} \text{Var}(\overline{X}) = \lim_{n\to\infty} \frac{1}{n} = 0$$

Thus $\overline{X}$ is consistent for μ_0.

3.4. Result 1: Probability limit Continuity

Given a continuous function $g(u)$, if $\hat{\theta} \xrightarrow{P} \theta_0$, then $g(\hat{\theta}) \xrightarrow{P} g(\theta_0)$.

If we want to estimate a function of the parameter $g(\theta_0)$ then $g(\hat{\theta})$ is a consistent estimator of $g(\theta_0)$.

Example:

Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu_0, \sigma_0^2)$.

Show that $\widetilde{S} = \sqrt{\widetilde{S^2}} = \sqrt{\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{n}}$ is consistent for $\sigma_0 = \sqrt{\sigma_0^2} = g(\sigma_0^2)$.

Note: we don't have $E(\widetilde{S})$ or $\text{Var}(\widetilde{S})$, but we do know $E(\widetilde{S^2})$ and $\text{Var}(\widetilde{S^2})$.

Then:

1. Show $\widetilde{S^2}$ is consistent for σ_0^2.
2. Apply result 1.

Recall:

$$E(\widetilde{S^2}) = \frac{(n-1)}{n}\sigma_0^2$$

$$\text{Var}(\widetilde{S^2}) = \frac{2(n-1)}{n^2}\sigma_0^4$$

Then:

$$\lim_{n\to\infty} E(\widetilde{S^2}) = \lim_{n\to\infty} \left(\frac{n-1}{n}\right) \sigma_0^2 = \sigma_0^2$$

So it is asymptotically unbiased.

$$\lim_{n\to\infty} \text{Var}(\widetilde{S^2}) = \lim_{n\to\infty} \frac{2(n-1)}{n^2}\sigma_0^4 = 0$$

Then $\widetilde{S^2}$ is a consistent estimator for σ_0^2.

Apply result 1:

The square root is a continuous function. Since $\widetilde{S}^2 \xrightarrow{P} \sigma_0^2$,

$$\widetilde{S} = \sqrt{\widetilde{S}^2} \xrightarrow{P} \sqrt{\sigma_0^2} = \sigma_0$$

So $\widetilde{S}$ is consistent for σ_0.

Notes:

- It is a lot easier to work with consistency than unbiasedness.

$$E[g(\hat{\theta})] \neq g[E(\hat{\theta})] \quad \text{but} \quad \text{plim}(g(\hat{\theta})) = g(\text{plim}(\hat{\theta}))$$

Example:

Let $X \sim N(\mu_0, \sigma_0^2)$.

Say we want to estimate $\tau_0 = e^{\mu_0}$.

We know $E(\overline{X}) = \mu_0$, but $E(e^{\overline{X}}) \neq e^{\mu_0}$.

On the other hand,

$$e^{\overline{X}} \xrightarrow{P} e^{\mu_0},$$

so $e^{\overline{X}}$ is consistent for e^{μ_0}.

Aside:

- Probability limit

$$\lim_{n\to\infty} P(|\hat{\theta} - \theta_0| < \epsilon) = 1 \quad \hat{\theta} \to \theta_0 \quad \text{or} \quad \text{plim}(\hat{\theta}) = \theta_0.$$

- Quadratic mean average

$$\lim_{n\to\infty} E[(\hat{\theta}-\theta_0)^2] = 0 \quad \text{iff} \quad \lim_{n\to\infty} \text{Bias}(\hat{\theta}) = 0 \quad \text{and} \quad \lim_{n\to\infty} \text{Var}(\hat{\theta}) = 0.$$

- Note:

$$\hat{\theta} \xrightarrow{q} \theta_0 \Rightarrow \hat{\theta} \xrightarrow{P} \theta_0.$$

  Quadratic mean implies consistency.

  So we really use the concept of quadratic mean to prove consistency.

3.4. Law of Large Numbers (LLN)

Sample averages play a key role in statistics.

- Most estimators are functions of sample averages.

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

Goal: Have a general theorem about the consistency of sample averages.

Weak Law of Large Numbers

Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim}$ from a distribution with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$.

Let $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then $\overline{X} \xrightarrow{P} \mu$ (Consistency).

Proof: By Chebyshev's inequality:

$$P(|\overline{X} - \mu| > \epsilon) \leq \frac{\text{Var}(\overline{X})}{\epsilon^2}$$

$$P(|\overline{X} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

$$\lim_{n\to\infty} P(|\overline{X} - \mu| > \epsilon) \leq \lim_{n\to\infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

So

$$\lim_{n\to\infty} P(|\overline{X} - \mu| > \epsilon) = 0$$

Therefore

$$\lim_{n\to\infty} P(|\overline{X} - \mu| < \epsilon) = 1$$

4. Convergence in Distribution

Goal: Approximate the distribution of estimators $\hat{\theta}$ as $n \to \infty$.

Definition: A sequence of random variables $\{X_n\}_{n=1}^{\infty}$ converges in distribution to a random variable Y if:

$$\lim_{n\to\infty} F_{X_n}(r) = F_Y(r) \quad \text{for } r \in \mathbb{R}$$

Notation: $X_n \xrightarrow{d} Y$

"The distribution of the sequence approaches the distribution of Y."

Notes:

- If $X_n \xrightarrow{P} Y \implies X_n \xrightarrow{d} Y$
- $X_n \xrightarrow{q} Y \implies X_n \xrightarrow{P} Y \implies X_n \xrightarrow{d} Y$

Goal: As before, have a general result about convergence in distribution of sample averages $\overline{X}$. Use Central Limit Theorem.

4.1.. Central Limit Theorem (CLT)

Lindeberg-Levy CLT

Let $(x_1, \ldots, x_n)$ be an iid sample from a distribution with $E(X) = \mu$ and $\text{Var}(x) = \sigma^2$. Let $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$. Then:

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} Y \quad \text{with } Y \sim N(0, \sigma^2)$$

or

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

or

$$\frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

(standardize by σ).

Notes:

1. Key result, but is only an approximation as $n \to \infty$.

2. There are many versions of CLT.

3. CLT is used a lot to approximate the distribution of many estimators.

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \sigma^2)$$

4. $\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$ is called the limiting distribution.

5. Based on CLT, the approximate distribution of $\overline{X}$ when n is large is:

$$\boxed{\overline{X} \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)}$$

or known as asymptotic distribution.

6. Before: If $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ then $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

7. Now: If $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim}$ from any pdf, then $\overline{X} \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$.

8. Note: The CLT is about $\overline{X}$, not about $(X_1, \ldots, X_n)$.

5. Useful Asymptotic Results

Let $a, b \in \mathbb{R}$

a) If

$$X_n \xrightarrow{P} a \quad \text{and} \quad Y_n \xrightarrow{P} b \quad \text{then}$$

$$X_n + Y_n \xrightarrow{P} a + b, \quad X_n \cdot Y_n \xrightarrow{P} a \cdot b$$

$$\frac{X_n}{Y_n} \xrightarrow{P} \frac{a}{b} \quad \text{if} \quad b \neq 0$$

b) **Slutsky's Theorem:** If $X_n \xrightarrow{d} U$ and $Y_n \xrightarrow{P} a$, then

i) $X_n \cdot Y_n \xrightarrow{d} aU$

ii) $X_n + Y_n \xrightarrow{d} U + a$

iii) $\frac{X_n}{Y_n} \xrightarrow{d} \frac{U}{a}$ if $a \neq 0$

c) **Delta Method:** Goal: Get limiting distribution of a function of an estimator. (See motivation and example from lecture.)

– **Theorem:** Let $\left\{\hat{\theta}_n\right\}_{n=1}^{\infty}$ be a sequence of estimators such that

$$\hat{\theta}_n \xrightarrow{P} \theta_0 \quad \text{and} \quad \sqrt{n}\left(\hat{\theta}_n - \theta_0\right) \xrightarrow{d} N(0, \Lambda)$$

Let $g(\theta)$ be a continuously differentiable function such that $g'(\theta_0) \neq 0$

$$\text{Then} \quad \sqrt{n}\left[g(\hat{\theta}) - g(\theta_0)\right] \xrightarrow{d} N(0, [g'(\theta_0)]^2 \Lambda)$$

6. Asymptotic Efficiency

Idea: When working with estimators $\hat{\theta}$ of θ_0, we usually get

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega).$$

Before: We got estimators that were UMVUE.

Goal: Get estimators that are "asymptotically UMVUE".

Definition: A sequence of estimators $\{\hat{\theta}_n\}_{n=1}^{\infty}$ is asymptotically efficient for θ_0 if:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1})$$

where $I(\theta) = E(S^2(\theta_0, X))$ or $\text{Var}(S(\theta_0, X))$ or $-E\left(\frac{\partial S(\theta_0, X)}{\partial \theta}\right)$.

Note:

- It's an extension of CRLB and UMVUE.

- For large enough n:

$$\hat{\theta} \overset{a}{\sim} N(\theta, [nI(\theta)]^{-1})$$

- Note $[nI(\theta)]^{-1} = \frac{1}{nI(\theta)} = \text{CRLB}$.

So, *an estimator $\widehat{\theta}$ is asymptotically efficient if*:

1. $\hat{\theta}^*$ is asymptotically unbiased.

2. Asymptotic variance of $\hat{\theta}^*$ equals CRLB.

Asymptotically efficient means it is impossible to find another asymptotically unbiased estimator with lower variance.

7. Asymptotic Properties of MLE

Theorem: (Asymptotic Properties of MLE) Let $(x_1, \ldots, x_n) \overset{\text{iid}}{\sim} f(x|\theta_0)$ and $\hat{\theta}_{\text{MLE}}$ be the MLE of θ_0.

Under some regularity conditions (see book):

a) $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0$ (MLE is consistent for θ_0).

b) $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$ (MLE is asymptotically normally distributed).

c) $\hat{\theta}_{\text{MLE}}$ is asymptotically efficient since $\hat{\theta}_{\text{MLE}} \overset{a}{\sim} N(\theta_0, [nI(\theta_0)]^{-1})$

(See final example from class.)